

# Night vision rendered darker and orange when nearing light sources with brightness on moody

MC-58177 Lighting

|          |                                                                                                                       |
|----------|-----------------------------------------------------------------------------------------------------------------------|
| Status   | Resolved / Fixed                                                                                                      |
| Affected | Minecraft 14w25b, Minecraft 14w26b, Minecraft 14w26c, Minecraft 14w27a, Minecraft 14w27b, Minecraft 14w30c (+26 more) |
| Fixed in | 19w45a                                                                                                                |
| Reported | 2014-06-20                                                                                                            |

## STEPS TO REPRODUCE

1. Create or enter a Creative world with default settings.
2. Open Options -> Video Settings.
3. Set the brightness to its minimum value.
4. Enter or create an area with light level 0.
5. Obtain the Night Vision effect.
6. Place a light source in the dark area.
7. Observe that blocks near, but not directly beside, the light source render darker and more orange, while the block directly beside it appears normal.

## OBSERVED BEHAVIOUR

## Actual behavior

With the in-game brightness set to **Moody**, applying **Night Vision** does not produce consistent lighting near light sources. In an otherwise light-level-0 area, blocks that are close to a newly placed light source - but not directly adjacent to it - appear significantly darker and more orange than expected. The block immediately next to the light source appears normally lit, creating an abrupt, inconsistent transition.

This effect also causes normally bright light-emitting blocks, such as **lava** and **fire**, to appear unusually dark. It is a visual rendering issue; the affected blocks' lighting levels and the light source's behavior are otherwise unchanged.

A related color shift occurs when the area is exposed to the sky at night. Under Night Vision, blocks beneath an opening to the night sky appear strongly blue compared with identical blocks in an enclosed, unexposed area. Fire and lava can also appear abnormally tinted or dark in this situation.

## EXPECTED BEHAVIOUR

With Night Vision active, blocks should remain consistently bright and retain their normal colors regardless of proximity to light sources, including lava and fire. Lighting transitions around placed or existing lights should not cause nearby blocks to become darker or orange while adjacent blocks appear normal. Likewise, areas exposed to the night sky should not develop an unusually blue tint compared with enclosed areas.

## ENVIRONMENT

Minecraft Java Edition Versions: 14w25b, 14w26b, 14w26c, 14w27a, 14w27b, 14w30c, 1.8, 1.8.1, 1.8.2-pre1, 1.10.2, 1.11, 1.12.1, 1.12.2, 17w43a, 18w22a, 1.13-pre1, 1.13-pre6, 1.13-pre10, 1.13, 18w30b, 1.13.1, 1.13.2, 18w48a, 18w48b, 18w49a, 19w07a, 1.14 Pre-Release 5, 1.14.2, 1.14.4, 19w35a, 19w42a, 19w44a (includes snapshots)

## ORIGINAL DESCRIPTION

## The bug

When exploring mineshafts using night vision, I found that the blocks nearing light sources always become darker and more orange. But the block just beside the light source is normal.

## How to reproduce

- Set brightness in video options to minimum

- Make/Go to a place with light level 0. (Note: It does something else if exposed to sky, but still looks weird though.)
- Get Night Vision effect.
- Place any kind of light source.
- The blocks near but not beside the light source are rendered darker. (Note: Lava blocks and Fire blocks are rendered too dark too!)

## New founds added 6/24:

Finally I got the idea why night vision looks weird under night sky. I found a ravine with a hole above it, and compare to the non exposed area, the place under the night sky are extremely blue.

## How to reproduce

- Set brightness in video options to minimum
- Make (since it's a little difficult to find) a huge (at least 30x10x30) room with light level 0. (Suggested that use stone with some lapis ores)
- Get Night Vision effect.
- Make sure it's night time in game.
- Break a hole on the roof.
- Enjoy! Don't forget to try Fire blocks, they look FUNNY!! (?! Just kidding XD)

## Code analysis

Code analysis by quadraxis can be found in this comment.

## MANUAL RESULT

|                  |                    |
|------------------|--------------------|
| Version used     |                    |
| Reproduced?      | YES / NO / PARTIAL |
| Crash file       |                    |
| Notes            |                    |
|                  |                    |
| Tested by / date |                    |

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-58177>